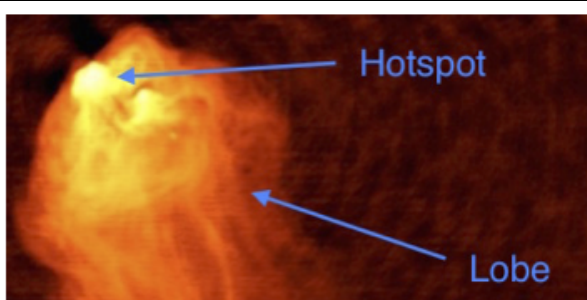
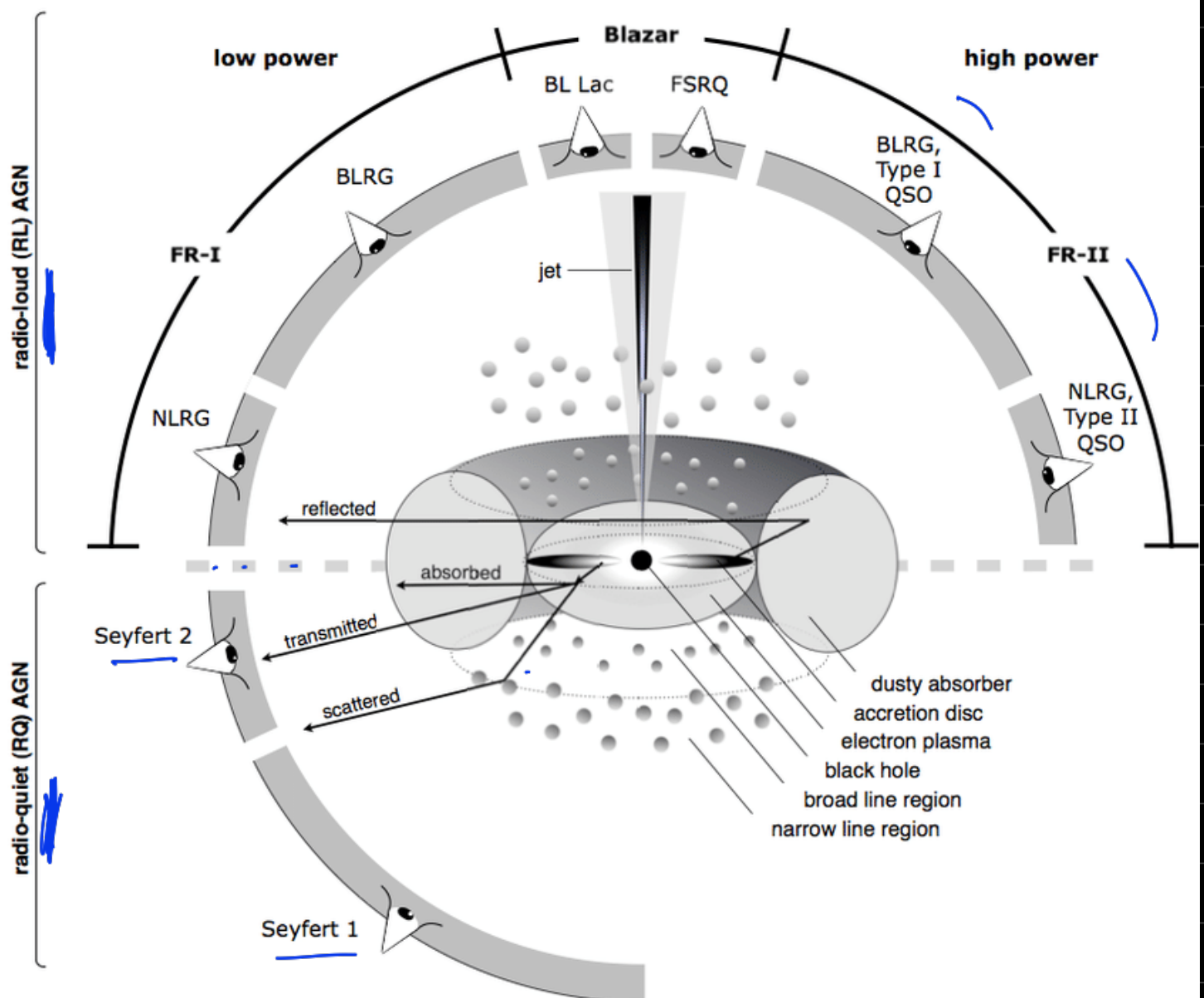
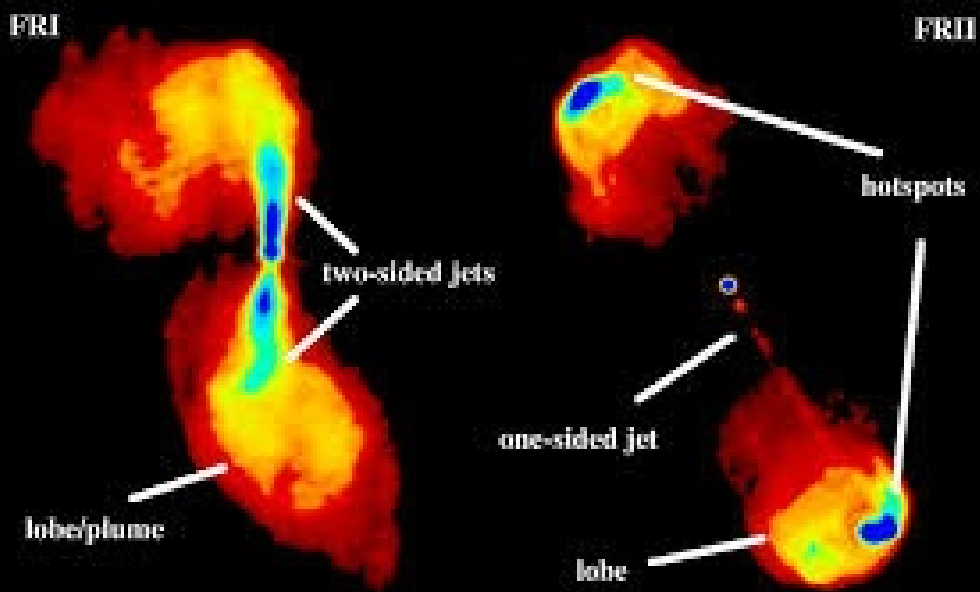
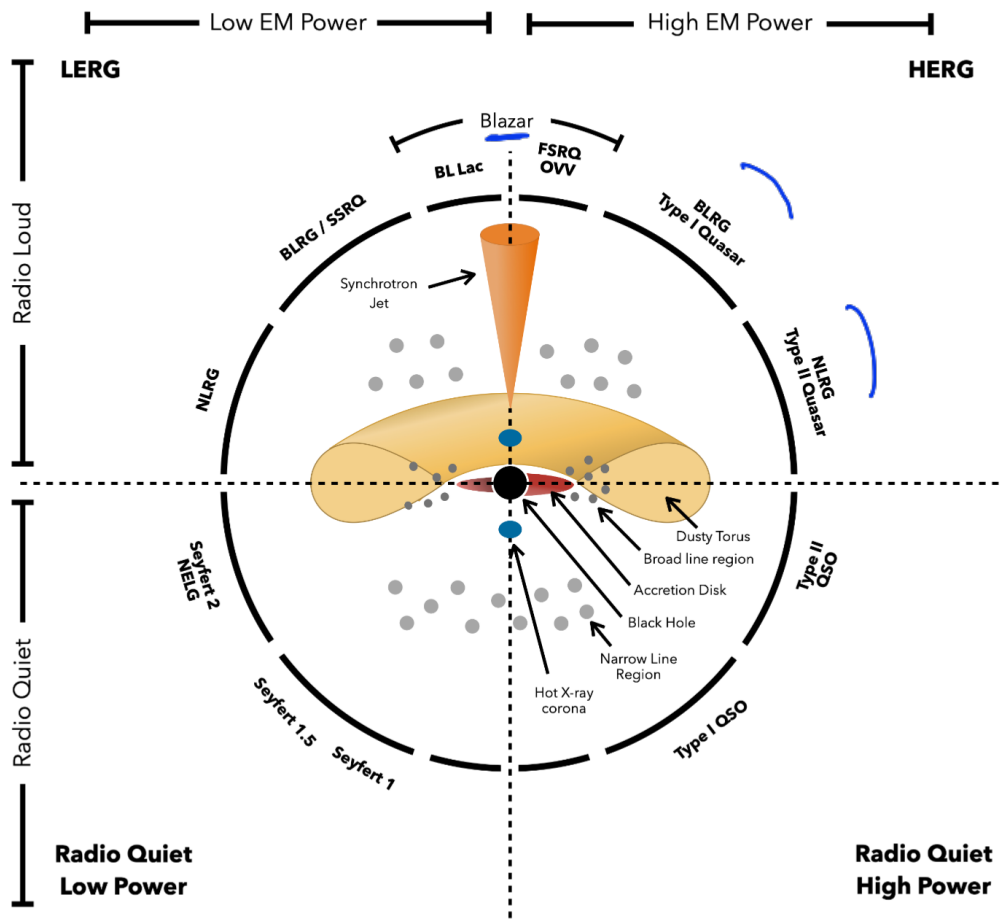
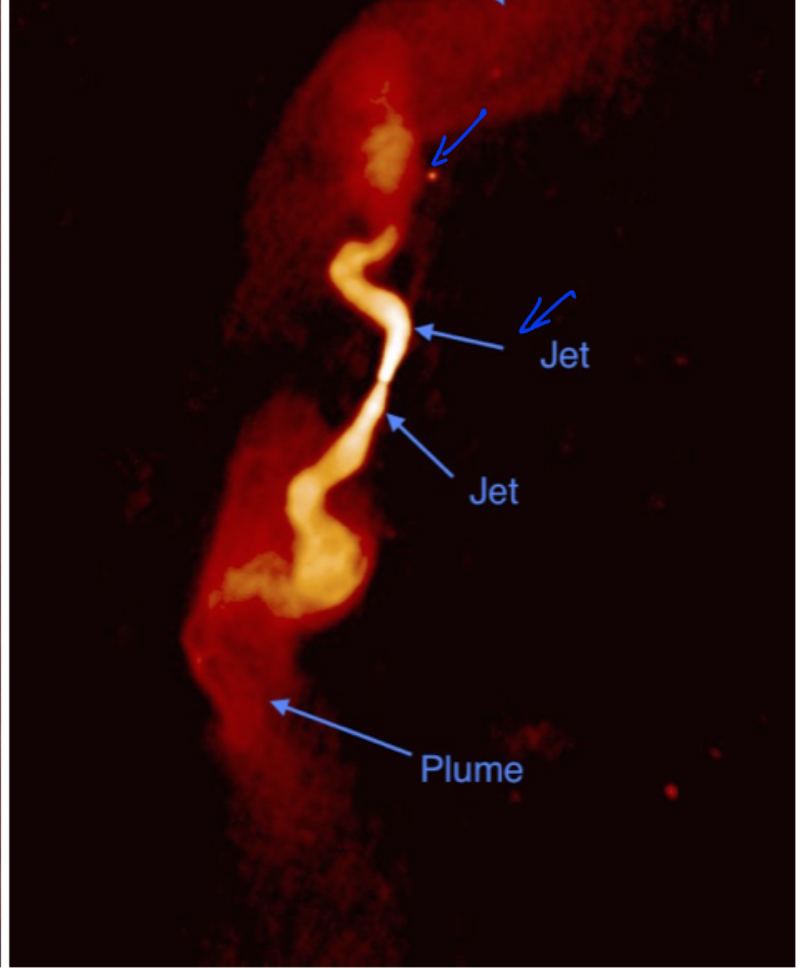
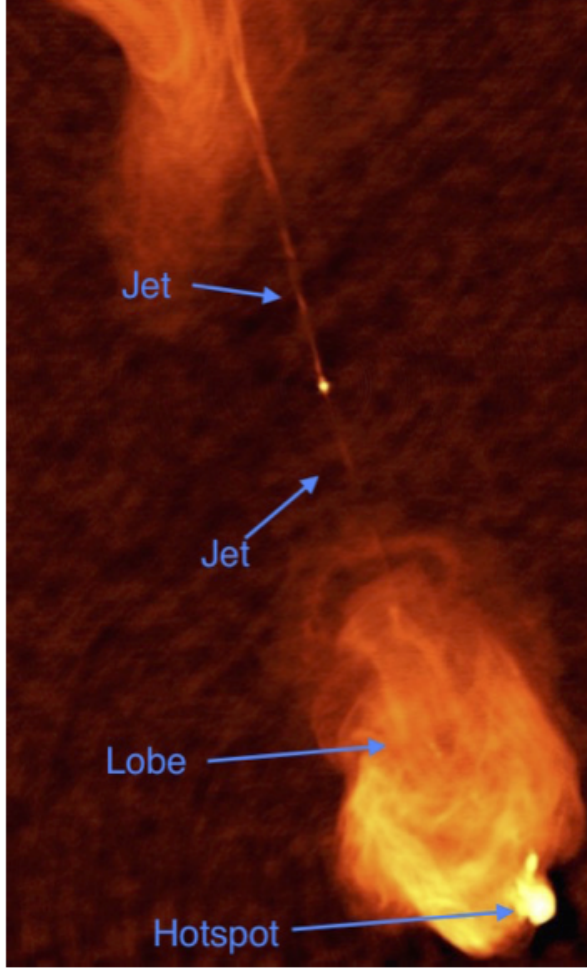






* Center of Galaxies:

AGN's: Active - Galactic Nuclei

→ Large amounts energy being emitted from the center

Classification:

→ Seyfert: $M_V \leq -23$

→ Quasars:

→ • Type I: Blue & UV emissions

• Type II: "narrow lines"

Blazars:

→ Radio Jets:

* AGN unification:

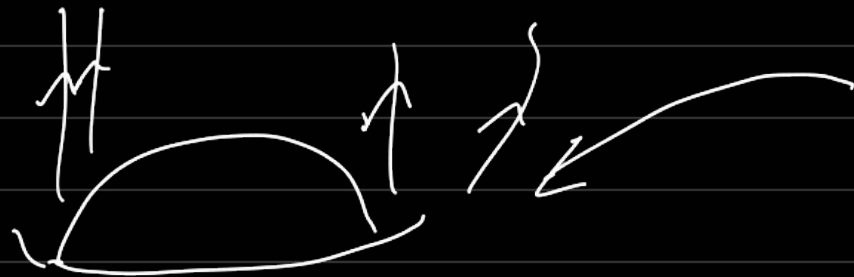
• Galaxy center CANNOT be Powered by supernova or nuclear fusion:

→ Eff: amount of energy we can get are not enough:

→ Models work for gravitational energy

⇒ Black holes

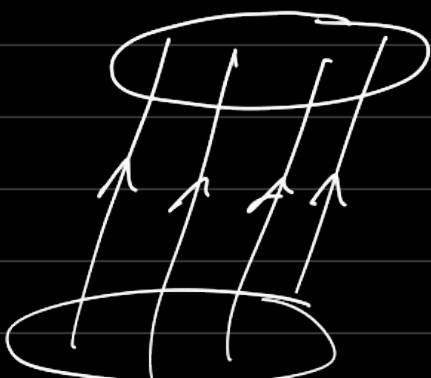
* Build model of a central BH:



• Highly ionized region ⇒ Strong EM fields

→ Look at the plasma physics:

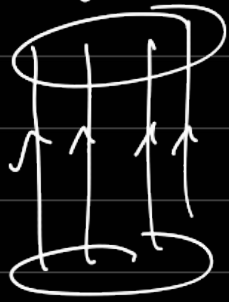
• Alfven flux freezing:



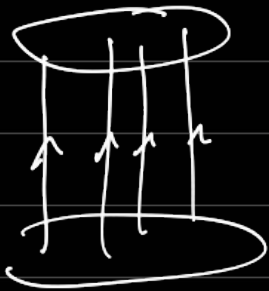
$$\frac{\partial \vec{B}}{\partial t} \sim \nabla \times (\underbrace{\vec{v} \times \vec{B}}_{\text{rel part}}) \quad \text{* Astro}$$

$$\text{Lab: } \frac{\partial \vec{B}}{\partial t} \sim \eta \nabla^2 \vec{B}$$

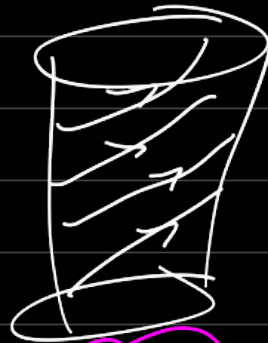
- Mag - fields act as a "solid"



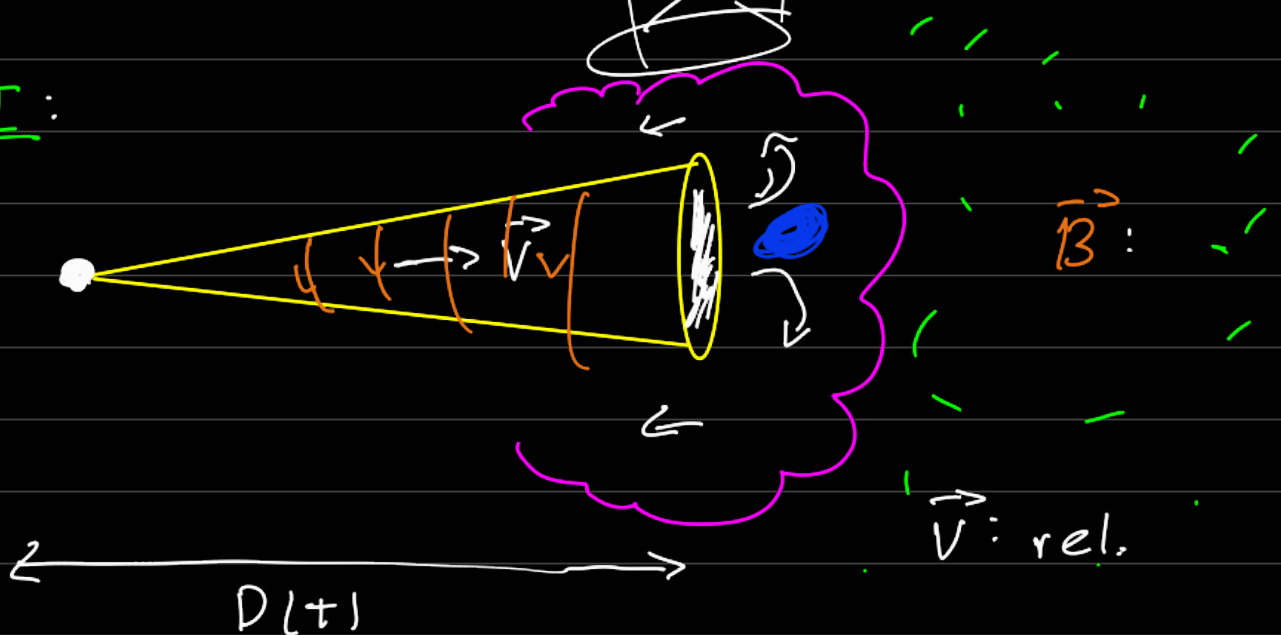
"Bend" →



"Twist" →



- FR II:



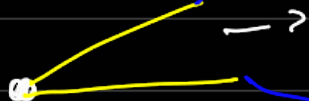
- Jet: collimated

- Lobe

- "Bright Spot"

- IGM:

FR I:



Jet:

tail: Diffuse tail

- Sensitive to IGM

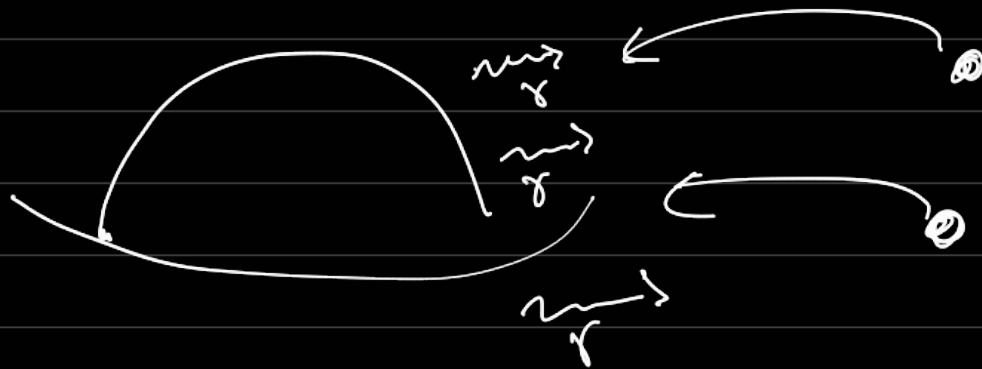
- Don't have as good a model

- Much more common / complex

Back to Nucleus:

Built a model from a BH: lets understand how that affects matter:

* Black Hole accretion:



* Understand why / how these AGNs are so luminous
→ Most den. objs in our obs. universe

* Requires GR:

$$\frac{2GM_{BH}}{c^2}$$

Sch. Radius:

$$R_{sch} = \frac{2GM_{BH}}{c^2}$$

- Look at GR, the smallest stable orbit occurs at $3R_s$.
- Vir thm as means to understand grav energy & its "conversion"
- Start w a particle out at ∞ and bring it to R :

$$\Delta U = 0 - \left(-\frac{GM_{BH}m}{R} \right) = \frac{GM_{BH}m}{R}$$
$$U \sim \frac{1}{R}$$
$$U(\infty) = \frac{1}{\infty} = 0$$

* Once its at R : assume we can 'get'

energy from this from its mass

$$E = \epsilon mc^2 \quad (\text{Energy we can "get"})$$

ϵ : radiative efficiency

Assume that all of E is from ΔU ,
then

$$\frac{GM_{BH}}{R} = \epsilon mc^2$$

$$\epsilon = \frac{GM_{BH}}{Rc^2}$$

$$(\epsilon_{max})_1 = \frac{1}{2}$$

• ϵ : for nuclear or supernovas: Does not work

Mechanism:

(grav Energy) $\longrightarrow$ (Heat) $\oplus$ (Radiation)

Vir theorem: $2K + U = 0$

$$\Delta K = \frac{1}{2} \Delta U$$

$$K \sim \frac{N}{2} kT$$

* Heat $\sim 50\%$ $\rightarrow$ Kinetic En.

Radiation $\sim 50\%$

$$E = \epsilon mc^2 = \frac{1}{2} \Delta U$$

• Better Bounds on ϵ

$$(\epsilon_{max})_2 = \frac{1}{4} = \frac{1}{2} (\epsilon_{max})_1 \quad *$$

• $R = R_s$

* Realistically: $\epsilon \approx 0.1$ * 10%.

Uses R_s
But we
use $3R_s$

* Look luminosities: $E = \epsilon mc^2$

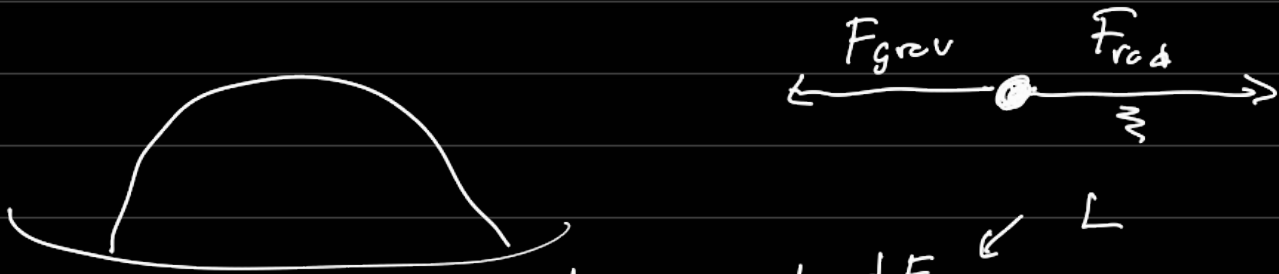
ϵ : rad. eff. | $\epsilon \approx 0.1$

$$L = \frac{dE}{dt} = \epsilon \dot{m} c^2$$

$$\dot{m} = \frac{L}{\epsilon c^2} *$$

* Not rel: we can also extract energy from a BH
 → Penrose process

• Maximum Accretion Lim: Eddington Limit:



$$E_r = pc : F_{\text{rad}} = \frac{dp}{dt} = \frac{1}{c} \frac{dE}{dt}$$

$$\tilde{F}_{\text{rad}} = \frac{1}{c} L_{\text{tot}}$$

$$F_{\text{rad}} = \left(\frac{\sigma_T}{4\pi R^2} \right) \tilde{F}_{\text{rad}} = \boxed{\frac{\sigma_T L}{4\pi R^2 c} = F_{\text{rad}}}$$

* F_{grav} : due to the mag fields: if we take e^- or p^+ : you have to take both
 • At best we can get H .

$$F_{\text{grav}} = \frac{G M_{\text{BH}} (m_e + m_p)}{R^2} \approx \frac{G M_{\text{BH}} m_p}{R^2}$$

→ Use New: Assume BH is "point source"

$$\underline{\text{Equate:}} \quad \frac{G M_{\text{BH}} m_p}{R^2} = \frac{\sigma_T L}{4\pi R^2 c}$$

Solve for L :

$$L_{\text{edd}} = \frac{G M_{\text{BH}} m_p 4\pi c}{\sigma_T}$$

use (Eddington Limit)

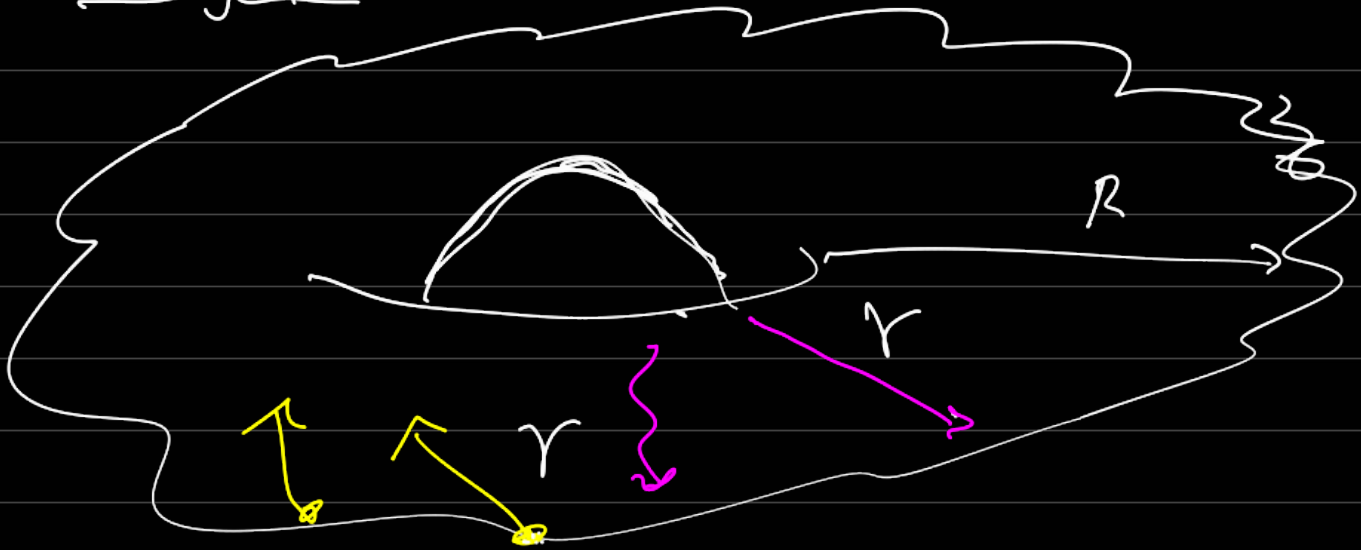
$$L_{\text{edd}} = 1.26 \times 10^{31} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) \text{ W}$$

* Obs:

$$\frac{M_{BH}}{M_\odot} \approx 0.001 \left(\frac{L_{bulge}}{L_\odot} \right)^{1.1}$$

• $L_{bulge} \approx L_{BH}$

* Crossing time:



$$R \leq \Delta \tau c$$

